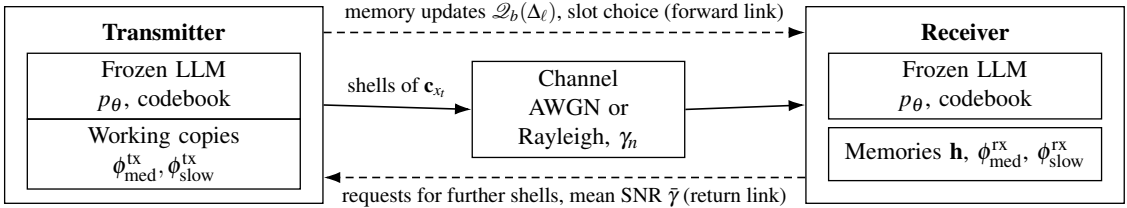
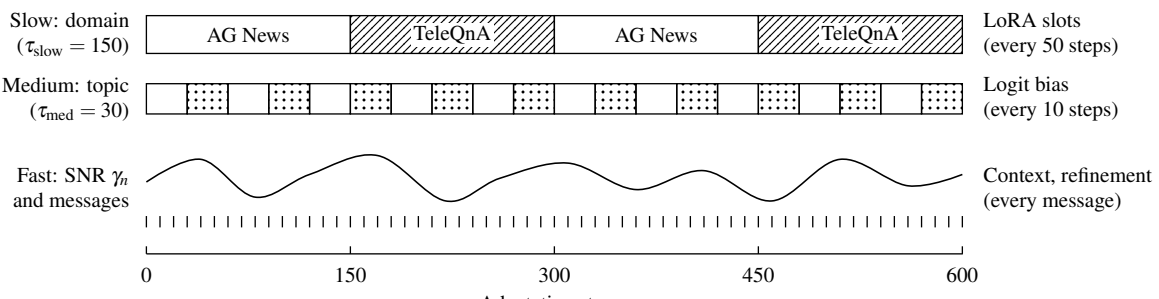
(a)


Transmitter
Frozen LLM
p_θ, codebook
Working copies
$\phi^{\text{tx}}_{\text{med}}, \phi^{\text{tx}}_{\text{slow}}$
memory updates $\mathcal{Q}_b(\Delta_\ell)$, slot choice (forward link)
shells of $\mathbf{c}_{x_t}$
Channel
AWGN or
Rayleigh, γ_n
requests for further shells, mean SNR $\bar{\gamma}$ (return link)
Receiver
Frozen LLM
p_θ, codebook
Memories $\mathbf{h}$, $\phi^{\text{rx}}_{\text{med}}$, $\phi^{\text{rx}}_{\text{slow}}$

(b)


Slow: domain
($\tau_{\text{slow}} = 150$)
AG News
TeleQnA
AG News
TeleQnA
LoRA slots
(every 50 steps)
Medium: topic
($\tau_{\text{med}} = 30$)
Logit bias
(every 10 steps)
Fast: SNR γ_n
and messages
Context, refinement
(every message)
0
150
300
450
600
Adaptation step n